



MACHINE LEARNING

Weekly Lecture Schedule • Fall 2026

Bachelor of Science in Information Technology • *From Hypotheses to Decisions* • every example drawn from IT project management

CREDIT HOURS	DURATION	PREREQUISITE	CONTENTS
3 (2–3): 2 lecture h + 3 lab h a week	16 teaching weeks; midterm Week 8, final after Week 16	Artificial Intelligence (HEC 2023 outline)	22 chapters in 6 parts, each with a runnable Python lab

📅	📖 Lecture topics	🔧 Laboratory (3 h)	📋 Assessment
1	1 What Machine Learning Is – precise definition, four paradigms, designing a learning system, learning as hypothesis search 2 Mathematics for Machine Learning – vectors, gradients, gradient descent, probability, maximum likelihood, entropy	A rule against a learned model; the chapter in NumPy	—
2	3 The Machine Learning Workflow and Toolkit – splitting data, preparing features, scikit-learn interface, pipelines and data leakage, baselines	A leak-proof pipeline	—
3	4 Concept Learning and the Version Space – general-to-specific ordering, Find-S, version spaces, candidate elimination, inductive bias	Candidate elimination in forty lines	Quiz 1 (Ch 1–3)
4	5 Linear Regression – mean squared error, least squares by formula, gradient descent, polynomial features, measuring a model	Least squares three ways	—
5	6 Logistic Regression – sigmoid and log-odds, decision boundary, cross-entropy, fitting by gradient descent, more than two classes	Logistic regression by hand and by library	Assignment 1 set
6	7 Overfitting, Bias–Variance and Regularisation – under- and overfitting, bias–variance decomposition, learning curves, regularisation, cross-validation	Find the bottom of the U	—
7	8 Measuring Classifier Performance – confusion matrix, precision and recall, ROC and PR curves, confidence intervals, McNemar’s test, class imbalance	Evaluate an imbalanced classifier	Assignment 1 due; Quiz 2 (Ch 4–7)
8	9 Decision Tree Learning – trees as hypotheses, entropy and information gain, gain ratio, pruning, inductive bias of ID3	ID3 by hand; pruning by cross-validation	Midterm (Ch 1–8)
9	10 Bayesian Learning and Naive Bayes – Bayes for hypotheses, Bayes optimal classifier, Laplace smoothing, Gaussian naive Bayes, text classification	Naive Bayes by counting and by library	—
10	11 Instance-Based Learning: k-Nearest Neighbours – distance and scale, choosing k , curse of dimensionality 12 Support Vector Machines – maximum margin, soft margin and hinge loss, the kernel trick	Neighbours, scaling and k ; margins and kernels	Assignment 2 set
11	13 Artificial Neural Networks – perceptron, linear separability and XOR, delta rule, backpropagation, activation functions 22 The Machine Learning Project – the project brief; choosing and scoping a problem, the plan	Backpropagation from scratch	Project set; proposal due
12	14 Ensemble Learning – committees of hypotheses, bagging, random forests, boosting, stacking	One tree against five ensembles	Assignment 2 due
13	15 Clustering – k -means, hierarchical agglomerative clustering, how many clusters 16 Dimensionality Reduction and Self-Organizing Maps – PCA, how many components, self-organizing maps	k -means and a dendrogram; PCA and a SOM	Quiz 3 (Ch 9–14)
14	17 Semi-Supervised Learning and the EM Algorithm – self-training, mixture models, EM with labelled and unlabelled data 18 Hidden Markov Models – Markov chains, forward algorithm, Viterbi, Baum–Welch, Monte Carlo estimation	EM by hand; forward, Viterbi, Baum–Welch	Project experiments frozen
15	19 Markov Decision Processes – Bellman equations, value iteration, policy iteration 20 Reinforcement Learning – Monte Carlo methods, exploration vs. exploitation, temporal-difference learning, Q-learning	Value and policy iteration; bandits and Q-learning	—
16	21 Responsible and Ethical Machine Learning – where bias enters, measuring fairness, privacy, explainability, accountability 22 Project presentations – the report and the presentation	Auditing a model for fairness	Project report and presentation

SUGGESTED ASSESSMENT WEIGHTS

10% Quizzes (three)	10% Assignments (two)	15% Project	25% Midterm, Ch 1–8	40% Final, whole course
------------------------	--------------------------	----------------	------------------------	----------------------------

I Foundations

IV Learning Without Full Labels

II Linear Models and Generalization

V Sequences, Decisions and Reinforcement

III Supervised Learning Algorithms

VI Responsible Practice

Chapter numbers refer to the Fall 2026 Machine Learning handout (Appendix F, Sixteen-Week Teaching Schedule). Chapters 7 and 8 are the hinge of the course and are taught before the algorithms on purpose. Weights are the handout’s suggested scheme, to be replaced by the department’s own. Set texts (HEC): T. M. Mitchell, *Machine Learning* (1997); K. P. Murphy, *Machine Learning: A Probabilistic Perspective* (2012).